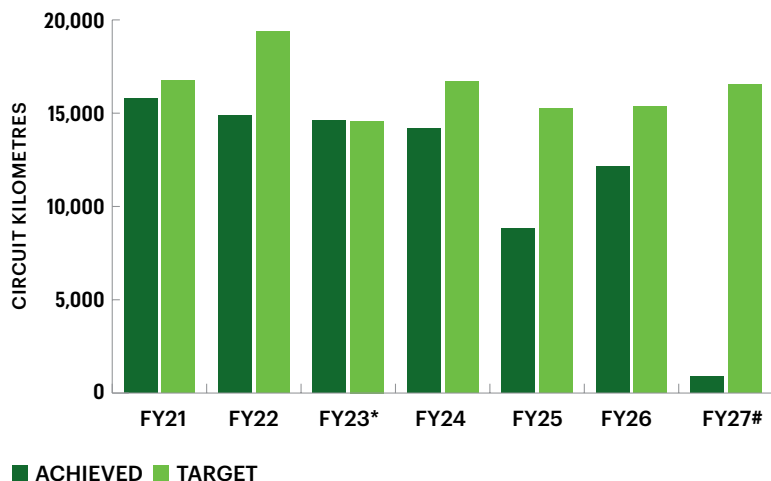


OUT OF LINE

- Transmission line targets remains unmet
- FY25 saw an almost 50% shortfall



NOTE *TARGET REVISED FROM 17,863 CIRCUIT KILOMETRES; #TILL APRIL 2026
SOURCE CENTRAL ELECTRICITY AUTHORITY

remain operationally and economically viable. For FY26-27, renewable energy projects in India are expected to face average connectivity delays of four-five months beyond their commissioning timelines. Around 20 GW out of 45 GW of projects could see delays extending beyond four months. This is most acute in Rajasthan's pooling stations. Over 12 GW of solar and wind capacity in Rajasthan, and around 8 GW in Western Region and Southern Region combined, face such delays. This not only creates financial stress for developers but also results in system-level opportunity costs due to unutilised generation capacity.

India's transmission network spans about 503,661 circuit kilometres (ckm). It comprises the national grid, the inter-state transmission system (ISTS), and the intra-state transmission system (InSTS), which operates within states and connects to the distribution system. The national grid connects India's five regional power grids: Northern, Western, Southern, Eastern, and North-eastern.

The latest National Transmission Plan (2022) seeks to expand the network to 648,190 ckm by FY32. This implies annual addition of around 24,000 ckm against the current average of 13,000 ckm added in the past five years. Meeting this goal will be critical to integrating India's target of 500 gigawatts (GW) of non-fossil capacity by 2030.

According to an analysis by Ember, over the past five years (FY2021-22 to FY2025-26), India has met about 80% of its transmission planning targets on average. Annual additions have been around 13,000 ckm, against a target of roughly 16,230 ckm across inter-state transmission system

of the problem," says Tata Power CMD Praveer Sinha.

Coal plants could not reduce generation beyond a point due to technical constraints, leaving limited room on the grid to absorb rising solar power. As a result, some solar generation had to be curtailed to maintain grid stability. Rapid growth in solar capacity is enabling solar to serve an increasing share of daytime demand, altering net load patterns that conventional generators have historically supplied. There is an economic cost at play as well. Operating thermal power plants at low capacity to allow higher injection of renewable energy into the grid during the day is expensive and leads to a higher per-unit cost of coal-fired electricity. Thermal plants also need to run at a minimum plant load factor to

INSTALLED SOLAR CAPACITY

- Solar addition is happening at a fast pace
- Solar to play a key role in India's energy transition targets

